The mathematics, step by step

Companion note to "What Survives a Placebo? Entropy, Market Coupling and the Plausibility of Stress Scenarios"

(version prepared for the *Journal of Financial Econometrics*)

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Abstract

This note works through all the mathematics of the journal version of the paper and of its online appendix, one step at a time. Compared with the note on the central-bank version, the section on inference is new: the bias of sample coupling (Bartlett's formula), why resampling p-values are exact, the volatility-preserving GARCH null, the within-year gap, the Monte Carlo check of size and power, Holm's multiple-testing procedure and the Clark–West test. It is written for a master's student who knows probability (expectation, density, multivariate normal) and linear algebra (determinant, trace, eigenvalues, positive definite matrices), but not necessarily information theory. Each result is stated, proved, illustrated with a numerical example, and linked to the exact place in the paper where it is used. Blue boxes give the intuition, orange boxes the link with the paper, red boxes the mistakes to avoid. Most sections end with a few solved exercises.

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The storyline on one page

The paper rests on a chain of seven ideas. Each gets its own section in this note.

1. **Entropy measures the uncertainty** of a distribution, in nats (section 1).
2. **The Kullback–Leibler divergence measures the gap between two distributions**, also in nats, and it does not depend on units (section 2).
3. **For a fixed mean and covariance, the Gaussian is the maximum-entropy distribution**: the "most ignorant" one. Any departure from the Gaussian is therefore information (section 3).
4. **The entropy of a return vector splits exactly into three pieces**: scale (volatilities), dependence (correlations) and shape (tails and skewness). Dropping scale gives the structural index $\mathcal{J}$, which is itself a divergence (sections 4 to 6).
5. **Receiving information lowers uncertainty on average** (a theorem), and we postulate that it then rebuilds slowly: hence a sawtooth dynamic (section 7; in the journal version these postulates are in the online appendix).
6. **But the estimator produces this sawtooth by itself.** Placebo nulls are therefore needed: one without dynamics, one that keeps volatility dynamics but not correlation dynamics. Only the coupling of markets under stress survives both (section 8).
7. **The nat becomes a unit of scenario severity**: k standard deviations cost $k^2/2$ nats, any scenario can be priced and the cheapest one found (section 9); and the coupling measure splits exactly by market segment, which gives the entropic CISS (section 10).

Notation. log is the natural logarithm. $X \in \mathbb{R}^N$ is a vector of returns with density p, mean μ and covariance Σ. We write $\Sigma = DRD$ with $D = \mathrm{diag}(\sigma_1, \ldots, \sigma_N)$ the volatilities and R the correlation matrix. $\mathcal{N}(\mu, \Sigma)$ is the normal distribution; q always denotes a Gaussian reference.

1 Entropy and its unit, the nat

1.1 The discrete case: counting questions

For a discrete variable taking values x_i with probabilities p_i, the *Shannon entropy* is

$$H(p) = -\sum_i p_i \log p_i.$$

A few values worth knowing: a fair coin has $H = \log 2 \approx 0.693$; a uniform distribution over M values has $H = \log M$; a certain variable has $H = 0$. Entropy is largest for the uniform distribution and zero for a distribution concentrated on one point.

In plain words

H measures "how much we do not know". $-\log p_i$ is the *surprise* of outcome i: a rare outcome is very surprising. Entropy is the average surprise.

The unit. Choosing the base of the logarithm *is* choosing the unit. With $\log_2$ we count in bits; with the natural logarithm, in *nats*. Since $\log_2 x = \log x / \log 2$, 1 nat $= 1/\log 2 \approx 1.443$ bits. A bit is the information that halves the set of possibilities; a nat divides it by $e \approx 2.718$. The whole paper is in nats.

1.2 The continuous case: differential entropy

For a density p on $\mathbb{R}^N$, define

$$h(p) = -\int p(x) \log p(x)\, dx = -\mathbb{E}_p[\log p(X)].$$

Example 1.1 (Uniform and normal distributions in one dimension)**.** The method is the same in both cases: write $\log p(x)$, then take the expectation of $-\log p(X)$.

Uniform on $[0, a]$**.**

1. The density is $p(x) = 1/a$ on $[0, a]$ and 0 elsewhere.
2. On $[0, a]$, $\log p(x) = \log(1/a) = -\log a$: a constant.
3. The expectation of a constant is the constant itself, so $h = -\mathbb{E}[\log p(X)] = -(-\log a) = \log a$.

Reading: the wider the interval, the larger the uncertainty. For $a = 2$, $h = \log 2 \approx 0.69$; for $a = 1$, $h = 0$; for $a = 1/2$, $h = -\log 2 < 0$ (more on this below).

Normal $\mathcal{N}(\mu, \sigma^2)$**.**

1. The density is $p(x) = \dfrac{1}{\sqrt{2\pi\sigma^2}} \exp\Big(-\dfrac{(x-\mu)^2}{2\sigma^2}\Big)$.
2. Take the logarithm. The log of a product is the sum of the logs, and $\log \exp(y) = y$:

$$\log p(x) = \log \frac{1}{\sqrt{2\pi\sigma^2}} - \frac{(x-\mu)^2}{2\sigma^2} = -\tfrac{1}{2}\log(2\pi\sigma^2) - \frac{(x-\mu)^2}{2\sigma^2}.$$

   (We used $\log(1/\sqrt{y}) = -\frac{1}{2}\log y$.)

3. Change the sign: $-\log p(x) = \frac{1}{2}\log(2\pi\sigma^2) + \dfrac{(x-\mu)^2}{2\sigma^2}$.
4. Take the expectation for $X \sim \mathcal{N}(\mu, \sigma^2)$. The first term is a constant and stays as it is. In the second, the only random part is $(X-\mu)^2$, whose expectation is *by definition* the variance: $\mathbb{E}[(X-\mu)^2] = \sigma^2$. Hence

$$h = -\mathbb{E}[\log p(X)] = \tfrac{1}{2}\log(2\pi\sigma^2) + \frac{\sigma^2}{2\sigma^2} = \tfrac{1}{2}\log(2\pi\sigma^2) + \tfrac{1}{2}.$$

5. Put the $\frac{1}{2}$ inside the logarithm by writing $\frac{1}{2} = \frac{1}{2}\log e$ (since $\log e = 1$), then $\frac{1}{2}\log A + \frac{1}{2}\log e = \frac{1}{2}\log(Ae)$:

$$\boxed{h = \tfrac{1}{2}\log(2\pi e\, \sigma^2) = \tfrac{1}{2}\log(2\pi e) + \log\sigma}$$

Reading: the mean μ has disappeared (shifting a distribution does not change its uncertainty), and only the volatility σ matters, through $\log\sigma$. Doubling volatility adds $\log 2 \approx 0.69$ nat. Numerically, for $\sigma = 1$, $h = \frac{1}{2}\log(2\pi e) = \frac{1}{2}\log(17.08) \approx 1.419$ nats.

Two important differences from the discrete case show up at once. First, h can be negative: the uniform on $[0, 1/2]$ has $h = -\log 2 < 0$. Second, and this is what matters for the paper, h **depends on units**.

Proposition 1.2 (Change of scale)**.** *For $a \neq 0$, $h(aX+b) = h(X) + \log|a|$. In dimension N, for invertible A, $h(AX+b) = h(X) + \log|\det A|$.*

Proof. If $Y = AX + b$, the change-of-variables formula gives $p_Y(y) = p_X(x)/|\det A|$ with $x = A^{-1}(y-b)$. Hence $h(Y) = -\mathbb{E}[\log p_Y(Y)] = -\mathbb{E}[\log p_X(X)] + \log|\det A| = h(X) + \log|\det A|$. □

Common pitfall

Expressing returns in per cent rather than decimals multiplies each coordinate by 100, and so adds $N \log 100$ to h. The *level* of h therefore has no absolute meaning; only its changes, and above all the *divergences* of the next section, do. This is why the paper builds all its indicators as divergences.

Exercise 1.3. Compute h for $X \sim \mathcal{N}(0, \sigma^2)$ with $\sigma = 1\,\%$, writing first $\sigma = 0.01$ and then $\sigma = 1$ (returns in per cent). Check that the gap is $\log 100 \approx 4.61$.

Solution. $h = \frac{1}{2}\log(2\pi e) + \log\sigma$, that is $1.419 - 4.605 = -3.186$, then 1.419. The gap is $\log 100$.

2 The Kullback–Leibler divergence

Definition 2.1. For two densities p and q (with $q > 0$ wherever $p > 0$),

$$D_{KL}(p\,\|\,q) = \int p(x) \log \frac{p(x)}{q(x)}\, dx = \mathbb{E}_p\left[\log \frac{p(X)}{q(X)}\right].$$

In plain words

It is the extra surprise, on average, when you believe the world follows q while it actually follows p. Another reading: when testing p against q, the log-likelihood ratio rises on average by $D_{KL}(p\,\|\,q)$ per observation. A divergence of one nat means that each observation multiplies the evidence in favour of p by e on average.

2.1 Three properties

Proposition 2.2 (Gibbs' inequality)**.** $D_{KL}(p\,\|\,q) \geq 0$*, with equality if and only if $p = q$ almost everywhere.*

The tool: Jensen's inequality. A function φ is *concave* if its graph always lies above its chords: for two points a, b and a weight $t \in [0,1]$, $\varphi(ta + (1-t)b) \geq t\varphi(a) + (1-t)\varphi(b)$. The logarithm is concave (its second derivative $-1/x^2$ is negative). Jensen extends this chord inequality to any average: for any random variable $Z > 0$,

$$\mathbb{E}[\log Z] \;\leq\; \log \mathbb{E}[Z] \qquad \text{(“the mean of the logs is below the log of the mean”).}$$

Example: Z equals 1 or 4 with probability $\frac{1}{2}$. Then $\mathbb{E}[\log Z] = \frac{1}{2}(0 + \log 4) = \log 2 \approx 0.69$, whereas $\log \mathbb{E}[Z] = \log 2.5 \approx 0.92$. Since the logarithm is *strictly* concave, equality holds only if Z is constant: if Z always equals c, both sides equal $\log c$.

Proof. We proceed in four steps.

Step 1: flip the fraction. Since $\log(a/b) = -\log(b/a)$,

$$-D_{KL}(p \,\|\, q) = -\mathbb{E}_p\left[\log \frac{p(X)}{q(X)}\right] = \mathbb{E}_p\left[\log \frac{q(X)}{p(X)}\right].$$

It is enough to show that this last number is ≤ 0.

Step 2: apply Jensen. Let $Z = q(X)/p(X)$, where X follows p. Jensen gives

$$\mathbb{E}_p\left[\log \frac{q(X)}{p(X)}\right] \;\leq\; \log \mathbb{E}_p\left[\frac{q(X)}{p(X)}\right].$$

Step 3: compute the expectation on the right. By definition of the expectation under p, we integrate against the density p, over the region where $p > 0$ (elsewhere X never falls):

$$\mathbb{E}_p\left[\frac{q(X)}{p(X)}\right] = \int_{p>0} p(x)\,\frac{q(x)}{p(x)}\,dx = \int_{p>0} q(x)\,dx.$$

The p's cancel: what remains is the mass q puts on the region where $p > 0$. This is a probability, so it is ≤ 1 (exactly 1 if q puts no mass outside this region).

Step 4: conclude. The logarithm is increasing, so $\log \int_{p>0} q \leq \log 1 = 0$. Chaining the steps,

$$-D_{KL}(p \,\|\, q) \;\leq\; \log \int_{p>0} q \;\leq\; 0, \qquad \text{that is} \qquad D_{KL}(p \,\|\, q) \geq 0.$$

Equality case. If $D_{KL}(p \,\|\, q) = 0$, both inequalities are equalities. Equality in Jensen requires $Z = q/p$ to be constant, say $q(x) = c\,p(x)$ wherever $p > 0$. Equality in step 4 requires $\int_{p>0} q = 1$, that is $c \int p = c = 1$. Hence $q = p$. ("Almost everywhere" only means that sets of probability zero, such as a few isolated points, can be ignored, since they change no integral.) □

Example 2.3 (Check on a simple case)**.** A biased coin $p = (0.9,\, 0.1)$ compared with a fair coin $q = (0.5,\, 0.5)$:

$$D_{KL}(p \,\|\, q) = 0.9 \log \frac{0.9}{0.5} + 0.1 \log \frac{0.1}{0.5} = 0.9 \times 0.588 + 0.1 \times (-1.609) = 0.529 - 0.161 = 0.368.$$

One of the two terms is negative, but the weighted sum is positive: exactly what the proposition guarantees. The negative term is always "dominated" because it carries the small weight 0.1.

Remark (A proof without Jensen). One can also use the elementary inequality $\log x \leq x - 1$ (the graph of the logarithm lies below its tangent at $x = 1$, with equality only at $x = 1$). With $x = q/p$: $\mathbb{E}_p[\log(q/p)] \leq \mathbb{E}_p[q/p - 1] = \int_{p>0} q - 1 \leq 0$. Same idea, written with a tangent instead of a chord.

Proposition 2.4 (Invariance under change of variables)**.** *If $Y = \phi(X)$ with ϕ bijective and differentiable, and p_Y, q_Y are the laws of Y when $X \sim p$ or $X \sim q$, then $D_{KL}(p_Y \,\|\, q_Y) = D_{KL}(p \,\|\, q)$.*

Proof. $p_Y(y) = p(x)/|J_\phi(x)|$ and $q_Y(y) = q(x)/|J_\phi(x)|$ with the *same* Jacobian. In the ratio p_Y/q_Y, the Jacobian cancels. □

Where this is used in the paper

This is the property that lets equity returns, rate changes and exchange rates sit in the same matrix (entropic CISS), and nats be compared across markets.

Not a distance. In general $D_{KL}(p\,\|\,q) \neq D_{KL}(q\,\|\,p)$, and the triangle inequality fails: hence the word "divergence". For example, for two centred Gaussians with variances σ_1^2 and σ_0^2, we show (section 9) that $D_{KL}(p_1\,\|\,p_0) = \frac{1}{2}(v^2 - 1 - 2\log v)$ with $v = \sigma_1/\sigma_0$. For $v = 2$ this gives 0.807, and for $v = 1/2$, 0.318.

2.2 The most useful case: two Gaussians with the same covariance

Proposition 2.5. *If $p_1 = \mathcal{N}(\mu_1, \Sigma)$ and $p_0 = \mathcal{N}(\mu_0, \Sigma)$, then $D_{KL}(p_1\,\|\,p_0) = \frac{1}{2}\,\Delta^\top \Sigma^{-1} \Delta$ with $\Delta = \mu_1 - \mu_0$.*

Proof. The normalising constants are identical and cancel. What remains is

$$\log \frac{p_1(x)}{p_0(x)} = -\tfrac{1}{2}(x-\mu_1)^\top \Sigma^{-1}(x-\mu_1) + \tfrac{1}{2}(x-\mu_0)^\top \Sigma^{-1}(x-\mu_0).$$

Under p_1, write $x - \mu_0 = (x - \mu_1) + \Delta$ and expand: the cross term $\Delta^\top \Sigma^{-1}(X - \mu_1)$ has zero expectation, and the two quadratic forms in $X - \mu_1$ cancel. What remains is $\frac{1}{2}\Delta^\top \Sigma^{-1}\Delta$. □

The quantity $\sqrt{\Delta^\top \Sigma^{-1} \Delta}$ is the *Mahalanobis distance*: the shift measured in standard deviations, taking correlations into account. In one dimension, $\Delta = k\sigma$ gives

k standard deviations cost $k^2/2$ nats, and η nats buy $\sqrt{2\eta}$ standard deviations.

A 3σ shock costs 4.5 nats; one nat buys $\sqrt{2} \approx 1.41$ standard deviations. This conversion rule runs through the whole paper.

3 Why the Gaussian is the zero point

Proposition 3.1 (Maximum entropy)**.** *Among all densities on $\mathbb{R}^N$ with mean μ and covariance $\Sigma \succ 0$, the Gaussian $q = \mathcal{N}(\mu, \Sigma)$ is the unique maximum-entropy density.*

What needs to be shown. Fix a mean and a covariance. Many distributions have them: a Gaussian, a suitably chosen uniform, a Laplace, a fat-tailed law… The proposition says that among all of them, the Gaussian has the highest entropy. To show it, we compare an arbitrary p with the Gaussian q with the same moments, using the divergence $D_{KL}(p\,\|\,q)$, which we already know is non-negative (Gibbs).

Ingredient 1: the log of a Gaussian is a degree-two polynomial. Start in one dimension. The density of $q = \mathcal{N}(\mu, \sigma^2)$ is $q(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\!\big(-\frac{(x-\mu)^2}{2\sigma^2}\big)$. The logarithm "undoes" the exponential (example 1.1):

$$\log q(x) = \underbrace{-\tfrac{1}{2}\log(2\pi\sigma^2)}_{\text{constant}} - \frac{(x-\mu)^2}{2\sigma^2}.$$

Expanding $(x-\mu)^2 = x^2 - 2\mu x + \mu^2$ shows it has the form

$$\log q(x) = a + b\,x + c\,x^2, \qquad a = -\tfrac{1}{2}\log(2\pi\sigma^2) - \frac{\mu^2}{2\sigma^2}, \quad b = \frac{\mu}{\sigma^2}, \quad c = -\frac{1}{2\sigma^2}.$$

This is what "$\log q$ is quadratic" means: a degree-two polynomial in x, with no x^3, x^4, etc. This property is *specific* to the Gaussian. For a Laplace law, $\log p(x) = \text{const} - |x|/b$ is not a polynomial; for a Student law, $\log p$ contains $\log(1 + x^2/\nu)$. In dimension N it is the same:

$$\log q(x) = -\tfrac{1}{2}\left[N \log 2\pi + \log\det \Sigma\right] - \tfrac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu),$$

a constant plus a quadratic form in x.

Ingredient 2: the expectation of a degree-two polynomial depends only on the first two moments. By linearity of expectation, for any variable X (with any law),

$$\mathbb{E}[a + bX + cX^2] = a + b\,\mathbb{E}[X] + c\,\mathbb{E}[X^2].$$

But $\mathbb{E}[X] = \mu$ and $\mathbb{E}[X^2] = \sigma^2 + \mu^2$ are fixed by the mean and variance. **All laws with mean μ and variance σ^2 therefore give the same value to** $\mathbb{E}[\log q(X)]$, Gaussian or not. This would not hold for a function containing x^4: its expectation would involve the kurtosis, which differs from one law to another.

Consequence. If p has the same mean and variance as q, p can be replaced by q in the expectation of $\log q$:

$$\mathbb{E}_p[\log q(X)] = \mathbb{E}_q[\log q(X)] = -h(q).$$

The last equality is the definition of the entropy of q.

Proof. Put the pieces together. By definition of the divergence, then of entropy,

$$D_{KL}(p \,\|\, q) = \mathbb{E}_p[\log p(X)] - \mathbb{E}_p[\log q(X)] = -h(p) - \mathbb{E}_p[\log q(X)].$$

Replace $\mathbb{E}_p[\log q(X)]$ by $-h(q)$ (consequence above):

$$D_{KL}(p \,\|\, q) = -h(p) + h(q) = h(q) - h(p).$$

By Gibbs, $D_{KL}(p \,\|\, q) \geq 0$, so $h(p) \leq h(q)$. Equality holds if and only if $D_{KL}(p \,\|\, q) = 0$, that is $p = q$: the Gaussian is the unique maximum. □

Example 3.2 (Numerical check, mean 0 and variance 1)**.**

- *Gaussian* $\mathcal{N}(0,1)$: $h = \frac{1}{2}\log(2\pi e) \approx 1.419$.
- *Uniform on* $[-\sqrt{3}, \sqrt{3}]$ (its variance is $(2\sqrt{3})^2/12 = 1$): $h = \log(2\sqrt{3}) \approx 1.242$.
- *Laplace* with density $\frac{1}{2b}e^{-|x|/b}$ and $b = 1/\sqrt{2}$ (variance $2b^2 = 1$): $h = 1 + \log(2b) \approx 1.347$.

The Gaussian does have the highest entropy. Let us also check ingredient 2 on the uniform: $\log q(x) = -\frac{1}{2}\log 2\pi - \frac{1}{2}x^2$, so $\mathbb{E}_p[\log q] = -\frac{1}{2}\log 2\pi - \frac{1}{2}\mathbb{E}_p[X^2] = -\frac{1}{2}\log 2\pi - \frac{1}{2} = -1.419 = -h(q)$, although p is not Gaussian. Finally $D_{KL}(p \,\|\, q) = 1.419 - 1.242 = 0.177$ nat: the "entropy deficit" of the uniform, what the Gaussian does not see.

In plain words

Fixing a mean and a variance, and assuming nothing else, leads to the Gaussian. Any other law with the same moments "knows more": it contains additional information (skewness, fat tails, nonlinear dependence). Measuring the departure from the Gaussian therefore measures this information.

Remark. One can also find the Gaussian with a Lagrangian: maximising $-\int p \log p$ subject to $\int p = 1$, $\int xp = \mu$, $\int xx^\top p = \Sigma + \mu\mu^\top$ gives a first-order condition in which $\log p$ is quadratic, hence p Gaussian. This route explains the form of the result but proves neither that it is a maximum nor that it is unique; the divergence proof does both in three lines.

Where does the variance come from? (the "bridge" of the online appendix). Maximum entropy takes the variance as given without saying where it comes from. The paper proposes measuring time in traded volume. Over an interval, let V be volume and Q net order flow. Two assumptions: (i) the change in log price is λQ (linear Kyle-type impact); (ii) given V, all we know about Q is $\mathbb{E}[Q \mid V] = m$ and $\mathrm{Var}(Q \mid V) = \varsigma^2 V$. Proposition 3.1 in one dimension then gives

$$\Delta X \mid V \sim \mathcal{N}(\lambda m,\ \lambda^2 \varsigma^2 V).$$

Unconditionally, ΔX is a *mixture* of Gaussians with random variances, hence fat-tailed. The moral: fat tails in calendar time do not contradict maximum entropy; they can come from a random clock. No empirical result depends on this bridge: it only justifies the Gaussian as the reference.

Exercise 3.3 (A mixture has fat tails)**.** Let $X = \sqrt{V}\, Z$ with $Z \sim \mathcal{N}(0,1)$ independent of $V > 0$. Show that the kurtosis of X is $3\,\mathbb{E}[V^2]/\mathbb{E}[V]^2 \geq 3$, with equality if and only if V is constant.

Solution. $\mathbb{E}[X^2] = \mathbb{E}[V]$ and $\mathbb{E}[X^4] = \mathbb{E}[V^2]\mathbb{E}[Z^4] = 3\mathbb{E}[V^2]$. The ratio is $\mathbb{E}[X^4]/\mathbb{E}[X^2]^2 = 3\mathbb{E}[V^2]/\mathbb{E}[V]^2$, and $\mathbb{E}[V^2] \geq \mathbb{E}[V]^2$ by Jensen (or because the variance of V is non-negative).

4 The entropy of a multivariate Gaussian

Proposition 4.1. *For $q = \mathcal{N}(\mu, \Sigma)$ on $\mathbb{R}^N$,*

$$h(q) = \tfrac{1}{2} \log\big((2\pi e)^N \det \Sigma\big).$$

Proof. Start from the expression of $\log q$ above: $h(q) = \frac{N}{2} \log 2\pi + \frac{1}{2} \log\det\Sigma + \frac{1}{2}\mathbb{E}[(X-\mu)^\top \Sigma^{-1}(X-\mu)]$. For the last expectation, use **the trace trick**: a scalar equals its trace, and the trace is invariant under cyclic permutation ($\mathrm{tr}(ABC) = \mathrm{tr}(CAB)$). Hence

$$\mathbb{E}\big[(X-\mu)^\top \Sigma^{-1}(X-\mu)\big] = \mathbb{E}\big[\mathrm{tr}\big(\Sigma^{-1}(X-\mu)(X-\mu)^\top\big)\big] = \mathrm{tr}\big(\Sigma^{-1}\Sigma\big) = \mathrm{tr}(I_N) = N.$$

So $h(q) = \frac{1}{2}\big[N \log 2\pi + \log\det\Sigma + N\big]$. □

Remark. The trace trick comes back several times. Remember: for any X with covariance Σ and any matrix A, $\mathbb{E}[(X-\mu)^\top A(X-\mu)] = \mathrm{tr}(A\Sigma)$, *whatever the law of X*. Only the first two moments enter.

4.1 Separating volatilities and correlations

With $\Sigma = DRD$, multiplicativity of the determinant gives $\det\Sigma = (\det D)^2 \det R = \big(\prod_i \sigma_i\big)^2 \det R$, so

$$h(q) = \underbrace{\tfrac{N}{2}\log(2\pi e)}_{\text{constant}} + \underbrace{\sum_i \log\sigma_i}_{\text{scale}} + \underbrace{\tfrac{1}{2}\log\det R}_{\text{dependence}}. \tag{1}$$

Proposition 4.2 (Correlation destroys entropy)**.** *For every correlation matrix R,* $\det R \le 1$*, with equality if and only if $R = I$. The dependence term is therefore always negative or zero.*

Proof. R is symmetric positive semi-definite, with eigenvalues $\lambda_1, \dots, \lambda_N \ge 0$ and $\sum_k \lambda_k = \operatorname{tr} R = N$ (the diagonal of R contains only 1s). By the arithmetic–geometric mean inequality,

$$\det R = \prod_k \lambda_k \le \Big(\tfrac{1}{N} \sum_k \lambda_k\Big)^N = 1,$$

with equality if and only if every λ_k equals 1, that is $R = I$. (This is a special case of Hadamard's inequality.) □

Where this is used in the paper

Equation (1) is the paper's "ceiling": what we would get if returns were Gaussian. It also explains why the scale channel is a nuisance: in a crisis, volatilities rise, so $h(q)$ *rises* (by 5.97 nats on the main panel), whereas we would like a crisis to show up as a *constraint*. Hence the idea of removing scale.

5 Mutual information and coupling

Definition 5.1. The mutual information of two vectors X and Y is $I(X;Y) = D_{KL}(p_{XY} \,\|\, p_X \otimes p_Y)$: the gap between the joint law and the law we would have if X and Y were independent. For N variables, the *total correlation* (or multi-information, Watanabe 1960) is $D_{KL}(p \,\|\, \prod_i p_i)$.

By Gibbs, $I \ge 0$, with equality if and only if the variables are independent. By proposition 2.4, I does not depend on units.

Proposition 5.2 (The Gaussian case)**.** *If $X \sim \mathcal{N}(0, \Sigma)$ with $\Sigma = DRD$, the total correlation is*

$$C(R) = D_{KL}\Big(\mathcal{N}(0,\Sigma) \,\|\, \mathcal{N}(0, D^2)\Big) = -\tfrac{1}{2} \log\det R.$$

For two variables with correlation ρ: $I = -\frac{1}{2}\log(1-\rho^2)$.

Proof. The marginals of a Gaussian are Gaussian, so $\prod_i p_i = \mathcal{N}(0, D^2)$. For a Gaussian vector, $D_{KL}(p \,\|\, \prod_i p_i) = \sum_i h(p_i) - h(p)$ (same argument as proposition 3.1: $\log \prod_i p_i$ is quadratic and p has the same variances). With (1): $\sum_i h(p_i) = \frac{N}{2}\log(2\pi e) + \sum_i \log \sigma_i$, and the difference with $h(p)$ is $-\frac{1}{2}\log\det R$. For $N = 2$, $\det R = 1 - \rho^2$. □

Example 5.3. For $\rho = 0.9$, $I = -\frac{1}{2}\log 0.19 = 0.83$ nat. For $\rho = 0.3$, $I = 0.047$ nat: mutual information rises very fast as ρ approaches 1.

Constant correlation. If every correlation equals ρ, $R = (1-\rho)I + \rho\,\mathbf{1}\mathbf{1}^\top$. The vector $\mathbf{1}$ is an eigenvector with eigenvalue $1 + (N-1)\rho$ (the "market mode"), and every vector orthogonal to $\mathbf{1}$ has eigenvalue $1 - \rho$. Hence

$$\det R = (1-\rho)^{N-1}(1 + (N-1)\rho), \qquad C = -\tfrac{1}{2}\big[(N-1)\log(1-\rho) + \log(1 + (N-1)\rho)\big].$$

For $N = 20$: $C = 1.97$ nats if $\rho = 0.26$ and $C = 2.69$ nats if $\rho = 0.32$. These are the average calm and stress correlations of the main panel; the gap, 0.72 nat, gives the order of magnitude of the 0.99-nat rise measured on the actual matrix.

Effective number of modes. A related spectral reading: with $\tilde{\lambda}_k = \lambda_k / N$ (which sum to 1, like probabilities), define $N^{\text{eff}} = \exp(-\sum_k \tilde{\lambda}_k \log \tilde{\lambda}_k)$. It is the exponential of the entropy of the spectrum: $N^{\text{eff}} = N$ if $R = I$, and $N^{\text{eff}} \to 1$ if a single mode carries all the variance. With the constant correlation above, N^{eff} goes from 14.6 to 12.8 (on the actual matrix, the paper finds 13.1 then 11.4).

In plain words

C and N^{eff} say the same thing in two ways: under stress, there are fewer independent sources of risk. But $C = -\frac{1}{2}\sum_k \log \lambda_k$ *weighs the small eigenvalues heavily* (the log blows up as $\lambda_k \to 0$). And these are the ones that collapse when assets couple. This is the explanation offered for the robustness of C compared with measures that look only at the largest eigenvalue (absorption ratio).

Exercise 5.4. Show that for small ρ, $C \approx \frac{1}{4}N(N-1)\rho^2$.

Solution. $\log(1-\rho) \approx -\rho - \rho^2/2$ and $\log(1+(N-1)\rho) \approx (N-1)\rho - (N-1)^2\rho^2/2$. The first-order terms cancel; at second order, $C \approx \frac{1}{2}\left[(N-1)\rho^2/2 + (N-1)^2\rho^2/2\right] = \frac{1}{4}N(N-1)\rho^2$. Consequence: C grows like N^2; comparing panels of different sizes requires normalisation.

6 The three channels and the structural index

6.1 The entropy deficit, or shape channel

Definition 6.1. Let q be the Gaussian with the same mean and covariance as the true law p. The *entropy deficit* is $\mathcal{D} = D_{KL}(p \,\|\, q)$.

Proposition 6.2. *$\mathcal{D} = h(q) - h(p) \geq 0$, with equality if and only if p is Gaussian.*

Proof. This is exactly the proof of proposition 3.1 read backwards: $\mathbb{E}_p[\log q] = -h(q)$ because $\log q$ is quadratic and p, q share the same moments, so $D_{KL}(p \,\|\, q) = -h(p) + h(q)$. □

Common pitfall

The identity $D_{KL}(p \,\|\, q) = h(q) - h(p)$ is false in general. It holds only because q is Gaussian *and* has the same first two moments as p.

6.2 The exact decomposition

Combining (1) and proposition 6.2 gives the central equation of the framework (eq. (1) of the paper):

$$h(p) = \frac{N}{2}\log(2\pi e) + \underbrace{\sum_i \log \sigma_i}_{\text{scale}} + \underbrace{\frac{1}{2}\log\det R}_{\text{dependence} = -C} - \underbrace{\mathcal{D}}_{\text{shape}}. \tag{2}$$

There is no approximation: it is an identity.

6.3 The structural index is a divergence

We remove the scale channel and keep the other two: $\mathcal{J} = \mathcal{D} + C$. This choice might look like a hack. It is not:

Proposition 6.3. *Let $q_i = \mathcal{N}(\mu_i, \sigma_i^2)$ be the Gaussians with the same means and volatilities as each asset. Then*

$$\mathcal{J} = \mathcal{D} + C = D_{KL}(p \,\|\, \textstyle\prod_i q_i) \;\geq 0.$$

Proof. $\log \prod_i q_i(x) = -\sum_i [\frac{1}{2}\log(2\pi\sigma_i^2) + \frac{(x_i-\mu_i)^2}{2\sigma_i^2}]$. Under p, $\mathbb{E}[(X_i-\mu_i)^2] = \sigma_i^2$, so

$$D_{KL}(p \,\|\, \textstyle\prod_i q_i) = -h(p) + \frac{N}{2}\log(2\pi e) + \sum_i \log \sigma_i.$$

Substitute (2) for $h(p)$: the constant and the scale term cancel, leaving $-\frac{1}{2}\log\det R + \mathcal{D} = C + \mathcal{D}$. □

Being a divergence gives $\mathcal{J}$ three properties for free: it is non-negative; it is zero if and only if returns are *Gaussian and independent* (a well-defined reference); and it depends neither on units nor on the level of volatility.

The chain rule. We also have the general identity

$$D_{KL}(p \,\|\, \textstyle\prod_i q_i) = \sum_i D_{KL}(p_i \,\|\, q_i) + D_{KL}(p \,\|\, \prod_i p_i)\,,$$

(marginal deficits + full multi-information). The paper estimates the first term asset by asset and replaces the second by its Gaussian part C. Nonlinear tail dependence is therefore left out: the estimated $\mathcal{J}$ *understates* the true divergence (a limit stated in the paper).

6.4 How the marginal deficit is estimated

For a standardised variable with skewness S and excess kurtosis K, close to Gaussian, the Edgeworth expansion gives

$$\mathcal{D} \approx \frac{S^2}{12} + \frac{K^2}{48}.$$

This helps understanding (a skewness channel and a tail channel, additive) but is unusable as an estimator on daily returns: the estimated K is dominated by a few extreme days, and squaring it amplifies the problem (the paper reports a factor-69 overstatement for a Student t with 4 degrees of freedom). Instead we use the Hyvärinen (1998) approximation, of the form

$$\mathcal{D} \approx k_1\big(\mathbb{E}[G_1(Z)]\big)^2 + k_2\big(\mathbb{E}[G_2(Z)] - \mathbb{E}[G_2(\nu)]\big)^2, \qquad \nu \sim \mathcal{N}(0,1),$$

where G_1 is a bounded odd function (it captures skewness) and G_2 a slowly growing even function (it captures tails). Since G_1, G_2 grow more slowly than any polynomial, the estimator is much less sensitive to extreme values.

An important property: $\mathcal{J}$ is even in skewness. The reference $\mathcal{N}(0,1)$ is symmetric, so replacing Z by $-Z$ does not change $D_{KL}(p_Z \,\|\, \varphi)$. Edgeworth shows it too: S enters squared. **A crash and its mirror image (a rally) have the same $\mathcal{J}$.** The index measures the *size* of information, never its direction. Hence the need for an outside postulate on the sign (next section).

7 Dynamics: a theorem and two postulates

7.1 The theorem: information lowers uncertainty on average

Let $h(X \mid Y = y)$ be the entropy of the conditional law of X given $Y = y$, and $\mathbb{E}_Y[h(X \mid Y)]$ its average over the values of the signal.

Proposition 7.1. $\mathbb{E}_Y[h(X \mid Y)] = h(X) - I(X;Y) \le h(X)$.

Proof. By definition, $p(x \mid y) = p(x,y)/p(y)$, so $\mathbb{E}_Y[h(X \mid Y)] = -\mathbb{E}[\log p(X,Y)] + \mathbb{E}[\log p(Y)] = h(X,Y) - h(Y)$. Moreover $I(X;Y) = \mathbb{E}\big[\log \frac{p(X,Y)}{p(X)p(Y)}\big] = h(X) + h(Y) - h(X,Y)$. Adding, $\mathbb{E}_Y[h(X \mid Y)] = h(X) - I(X;Y)$, and $I \ge 0$ by Gibbs. □

Common pitfall

The theorem holds *on average*. A particular realisation of the signal can increase entropy. Example: X equals a with probability 0.99, so $H(X) = 0.056$ nat. With probability 0.98 the signal removes all doubt ($H = 0$); with probability 0.02 it makes the two outcomes equally likely ($H = \log 2 = 0.693$, twelve times more than before). The average, $0.02 \times 0.693 = 0.014$, is indeed below 0.056. This is why the framework predicts jumps that are *mainly*, not exclusively, one-sided.

Example 7.2 (A noisy Gaussian signal)**.** Let $X \sim \mathcal{N}(0, \sigma^2)$ be tomorrow's return and $Y = X + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ independent. The pair (X, Y) is Gaussian with $\rho^2 = \sigma^2/(\sigma^2 + \sigma_\varepsilon^2)$, so by proposition 5.2

$$I(X;Y) = -\tfrac{1}{2}\log(1 - \rho^2) = \tfrac{1}{2}\log\Big(1 + \frac{\sigma^2}{\sigma_\varepsilon^2}\Big).$$

With $\sigma = 1\,\%$: a sharp signal ($\sigma_\varepsilon = 0.5\,\%$) brings 0.80 nat, an average signal ($1\,\%$) 0.35 nat, a rumour ($3\,\%$) 0.05 nat. A factor of fifteen separates clear news from a rumour.

7.2 Postulate 1: relaxation (Ornstein–Uhlenbeck process)

Without news, we postulate that $\mathcal{J}$ reverts to its normal level $\bar{\mathcal{J}}$:

$$d\mathcal{J}_t = -\kappa(\mathcal{J}_t - \bar{\mathcal{J}})\,dt + \sigma_{\mathcal{J}}\,dW_t.$$

The half-life. Taking expectations, the Brownian term vanishes and $m(t) = \mathbb{E}[\mathcal{J}_t] - \bar{\mathcal{J}}$ satisfies $m' = -\kappa m$, so $m(t) = m(0)e^{-\kappa t}$. The gap halves when $e^{-\kappa t} = 1/2$, that is

$$t_{1/2} = \frac{\ln 2}{\kappa}.$$

In discrete time (daily data), the model becomes an AR(1), $\mathcal{J}_{t+1} - \bar{\mathcal{J}} = \phi(\mathcal{J}_t - \bar{\mathcal{J}}) + \varepsilon_t$ with $\phi = e^{-\kappa}$, and the half-life is $\ln 2/(-\ln \phi)$ days. This is how the 121-day half-life (against 92 under the null) is estimated.

Why this process? Doob's theorem (1942) says that the Ornstein–Uhlenbeck process is the only continuous process that is Markovian, Gaussian and stationary. If we want a process with no memory beyond the current state, a long-run level, and noise about which we assume nothing beyond its variance (maximum entropy again), there is no choice. We then add the jumps caused by news:

$$d\mathcal{J}_t = -\kappa(\mathcal{J}_t - \bar{\mathcal{J}})\,dt + \sigma_{\mathcal{J}}\,dW_t + dZ_t,$$

where $Z_t = \sum_{j:\,t_j \le t} z_j$ is the sum of positive jumps up to t (Barndorff-Nielsen and Shephard, 2001, class). What remains arbitrary: constant κ, linear mean reversion, and the fact that the Brownian term could make $\mathcal{J}$ negative.

7.3 Postulate 2: the sign

Since $\mathcal{J}$ does not see direction (section 6), we add a minimal assumption: jumps come with negative skewness, justified by the leverage effect (falls raise volatility more than rises of the same size). This is hypothesis H5.

7.4 Why dependence and tails should rise together

Major news hits every asset on the same day. On that day, they move together (coupling C rises) and each records an unusual loss (its deficit $\mathcal{D}$ rises). The two channels describe the same day seen from two places: the cross-section and each asset's time series. This is hypothesis H6, which we will see the null reproduces.

8 Inference: why and how the placebo nulls work

This is the most important section for understanding the results of the journal version. It explains why estimated entropy measures are biased and move on their own, how a resampling null corrects for it, why a second, volatility-preserving null is needed for claims about dependence, and how the size and power of the test are checked.

8.1 The bias of a convex function of estimated moments

Lemma 8.1 (Jensen applied to an estimator)**.** *If $\hat{\theta}$ is an (approximately) unbiased estimator of θ and f is convex, then $\mathbb{E}[f(\hat{\theta})] \ge f(\theta)$.*

$C = -\frac{1}{2}\log\det R$ is a convex function of R ($-\log\det$ is convex on positive definite matrices), and $\mathcal{D}$ is non-negative and zero only at the Gaussian point. Both are therefore **biased upward**: estimation noise looks like information.

8.2 How large is the bias of coupling? (Proposition 3 of the paper)

Take N *independent* Gaussian assets, so the true coupling is $C = 0$, and T observations. The likelihood-ratio statistic for testing $H_0 : R = I$ ("sphericity" of the correlation matrix) is

$$\Lambda = -T\log\det\hat{R}.$$

Classical asymptotic theory says that $-2\log(\text{likelihood ratio})$ is approximately χ^2 with as many degrees of freedom as free parameters restricted by H_0. Here H_0 fixes the $N(N-1)/2$ off-diagonal correlations, so $\Lambda \approx \chi^2_{N(N-1)/2}$. Bartlett (1950) showed that replacing T by $T' = T-1-(2N+5)/6$ makes the approximation much better in small samples. Since a χ^2_k variable has mean k,

$$\mathbb{E}\big[-\log\det \hat{R}\big] \approx \frac{N(N-1)}{2T'} \quad \Longrightarrow \quad \mathbb{E}\big[C(\hat{R})\big] \approx b(N,T) = \frac{N(N-1)}{4(T-1-(2N+5)/6)}.$$

Example 8.2 (The paper's panels)**.** For $N = 20$, $T = 252$: $T' = 251 - 7.5 = 243.5$ and $b = 380/974 = 0.39$ nat. For $N = 48$, $T = 252$: $T' = 251 - 16.8 = 234.2$ and $b = 2256/936.8 = 2.41$ nats. The simulation of the paper (table 2) gives 0.39 nat for $N = 20$: the formula is accurate. The bias is as large as a third of the effect we want to measure on stocks (0.99 nat).

In plain words

Why fourth moments barely matter: for two independent variables with finite fourth moments, $\sqrt{T}\hat{\rho} \to \mathcal{N}(0,1)$ whatever their marginal laws, because $\mathrm{Var}(XY) = \mathrm{Var}(X)\,\mathrm{Var}(Y)$ when X and Y are independent. The bias formula only uses the variance of $\hat{\rho}$, so it holds for Student-t_5 data too (the simulation gives 0.39).

Two practical consequences. (1) Coupling levels on panels of different size are not comparable: subtract $b(N,T)$ first. The bias-corrected effective number of modes is $N\exp\{-2[C(\hat{R}) - b]/N\}$. (2) A *difference* between two windows of the same length (stress minus calm) has the bias removed to first order. This is why every test in the paper is on a gap, not on a level.

Common pitfall

Shrinkage (Ledoit–Wolf) pulls $\hat{R}$ towards I and therefore removes most of the bias (0.002 nat in the simulation). It does not remove the *dynamic* artefact of the next subsection.

8.3 The mechanical sawtooth

When an extreme day enters the window, the estimated moments jump, and a convex function of these moments jumps *upward whatever the sign of the day.* As the day ages (exponential weighting) or leaves the window, the index decays slowly.

A short calculation. Take a window of T standardised returns whose sample correlation is exactly I (independent assets), and add one day on which every asset moves by the same amount a standard deviations. Ignoring the change in the means, the new second-moment matrix is $I + x\mathbf{1}\mathbf{1}^\top$ with $x = a^2/T$, and the new correlation matrix is $R' = (I + x\mathbf{1}\mathbf{1}^\top)/(1+x)$. By the matrix determinant lemma, $\det(I + x\mathbf{1}\mathbf{1}^\top) = 1 + Nx$, so

$$\Delta C = -\tfrac{1}{2}\log\det R' = \tfrac{1}{2}\big[N\log(1+x) - \log(1+Nx)\big] \approx \frac{N(N-1)}{4}\,x^2 \quad (x \text{ small}).$$

The first-order terms cancel and what remains is positive: a single common day raises coupling, whatever its sign. For $N = 20$, $T = 252$ and a 10-standard-deviation day ($x = 0.40$), $\Delta C =$

$\frac{1}{2}[20\log 1.40 - \log 8.94] = 2.2$ nats. The day then leaves the estimate only at the rate at which the window forgets: with exponential weights of half-life 60 days, its weight halves every 60 days.

On i.i.d. Gaussian data the paper's pipeline finds 37 positive jumps of $\mathcal{J}$, no negative one, and an apparent half-life of 35 days, with no event at all.

8.4 The resampling test and why its p-value is valid

Construction. Let $S(X)$ be a statistic computed by the *whole* pipeline on the $T \times N$ panel X (rolling estimation, stress regime, jump detection, gap). A null is a way of generating panels X^* that keep some features of X and destroy others. We draw $B = 200$ of them and compute $S(X_1^*), \ldots, S(X_B^*)$.

The p-value. In the predicted direction,

$$\hat{p} = \frac{1 + \#\{b : \ S(X_b^*) \geq S(X)\}}{B + 1}.$$

If, under the null hypothesis, the observed panel is just one more draw from the same mechanism, then $S(X), S(X_1^*), \ldots, S(X_B^*)$ are *exchangeable*: the observed value is equally likely to have any rank among the $B + 1$ values. The probability that it ranks in the top k is $k/(B + 1)$, so $P(\hat{p} \leq \alpha) \leq \alpha$ exactly, without any asymptotic argument. The "+1" is what makes this exact; with $B = 200$ the smallest possible p-value is $1/201 \approx 0.005$.

Common pitfall

Reporting $z = (S(X) - \bar{S}^*)/s_{S^*}$ ("six standard deviations above the null") is a useful effect size, but it is not a p-value: the null distribution need not be normal. The paper reports $\hat{p}$ for inference and z only as a description.

8.5 Null 1: i.i.d. resampling of days

Draw rows of X with replacement. Each row (one day, N returns) is intact, so the null keeps each asset's fat tails, each day's cross-sectional correlation and common extreme days. It destroys *all* dynamics: there is no volatility clustering, no persistence, no crisis that lasts. It answers the question "beyond the fact that turbulent periods contain big days, does the measure detect any dynamics?"

8.6 Null 2: constant-correlation GARCH (filtered historical simulation)

Why a second null. Under the i.i.d. null, a "stress" period is a random cluster of big days. In real data, stress comes with volatility clustering: high-volatility days follow each other for weeks. Could volatility clustering alone, with correlations that never change, produce the coupling gap? Two mechanisms make this plausible. Forbes and Rigobon (2002) showed that measured correlation rises mechanically when the variance of a common factor rises. And our stress regime is defined by high volatility, so it selects precisely the windows where this happens.

Construction. For each asset i, fit a GJR-GARCH(1,1):

$$r_{i,t} = \mu_i + \varepsilon_{i,t}, \qquad \varepsilon_{i,t} = \sqrt{h_{i,t}}\, z_{i,t}, \qquad h_{i,t} = \omega_i + (\alpha_i + \gamma_i \mathbf{1}\{\varepsilon_{i,t-1} < 0\})\varepsilon_{i,t-1}^2 + \beta_i h_{i,t-1}.$$

The term γ_i makes volatility react more to falls than to rises (the leverage effect). Estimate by maximising the Gaussian likelihood (quasi-MLE, consistent even if z is not Gaussian). Compute the standardised residuals $\hat{z}_{i,t} = \hat{\varepsilon}_{i,t}/\sqrt{\hat{h}_{i,t}}$ and store them as a $T \times N$ matrix $\hat{Z}$.

To simulate: draw rows of $\hat{Z}$ with replacement (as in null 1), then rebuild each asset's returns by running its own GARCH recursion forward with the drawn shocks. The result has three properties:

1. each asset keeps its own volatility clustering and leverage effect;
2. all assets receive the shocks of the same historical day, so a common large day still hits everyone, and volatility bursts are common;
3. the correlation of the shocks z_t is the same every day: the *conditional* correlation is constant (Bollerslev's CCC model).

So volatility moves as in a GARCH world and correlations do not. A coupling gap beyond this null is a genuine dependence dynamic.

In plain words

Null 1 says: "it is not just big days". Null 2 says: "it is not just volatility". The 21-day block null that the earlier version rejected kept *both* volatility and correlation dynamics over a month, i.e. the crisis itself. Null 2 keeps the volatility part only, which is exactly the counterfactual the coupling claim needs.

8.7 A trend is not a stress effect: the within-year gap

Correlations between stocks rose over 1990–2022, and stress dates cluster in a few years (2008, 2020). If coupling simply trends up and stress dates happen to be late, the stress-minus-calm gap picks up the trend. The fix is a regression with year fixed effects:

$$C_t = \theta\, \mathbf{1}\{t \in \text{stress}\} + \sum_y \delta_y \mathbf{1}\{t \in \text{year } y\} + u_t.$$

By the Frisch–Waugh–Lovell theorem, $\hat{\theta}$ is the slope of the regression of the within-year demeaned C_t on the within-year demeaned stress dummy: it compares stress and calm dates *of the same year*. The paper finds that $\hat{\theta}$ is about 61% smaller than the raw gap, but still beyond both nulls.

8.8 Checking size and power by Monte Carlo

A test has the right *size* if, when the null hypothesis is true, it rejects in about 5% of samples at the 5% level; its *power* is the rejection rate when the hypothesis is false. To check both, simulate panels from known processes, apply the whole test to each, and count rejections:

- i.i.d. Student-t, constant correlation: no dynamics at all. Both nulls should reject about 5%.
- GARCH with constant correlation: volatility clustering, no correlation dynamics. A good test of *dependence* should still reject about 5%. The i.i.d. null rejects 17% of the time, the GARCH null 6.5%.

- Markov-switching volatility, constant correlation: same question with regime-type volatility.
- Markov-switching volatility *and* correlation (0.30 calm, 0.50 in stress): the effect is real; rejection rates measure power (68% for the GARCH null).

Each null fails somewhere. The i.i.d. null over-rejects under GARCH clustering (17%); the GARCH null over-rejects when volatility switches regimes (11%), because a smooth GARCH recursion cannot copy abrupt switches. The paper therefore asks a result to pass *both* nulls. Under this joint rule the worst rejection rate over the three null processes is 6.5% at the 5% level and 1.0% at the 1% level, while the power against the correlation-switching process stays at 46%. With 200 simulations, a rate of 6.5% is within about one standard error ($\sqrt{0.05 \cdot 0.95/200} \approx 1.5$ points) of 5%.

8.9 Testing many hypotheses at once: Holm's procedure

The verdict table tests m statistics. If each is tested at 5%, the chance of at least one false rejection grows quickly with m. The Holm (1979) procedure controls this *family-wise error rate*. Sort the p-values $p_{(1)} \leq \cdots \leq p_{(m)}$; reject $H_{(1)}$ if $p_{(1)} \leq \alpha/m$, then $H_{(2)}$ if $p_{(2)} \leq \alpha/(m-1)$, and so on, stopping at the first failure. Equivalently, the adjusted p-value of the k-th smallest is $\max_{j \leq k} \min\{1, (m-j+1)p_{(j)}\}$. Holm is always at least as powerful as Bonferroni (which uses α/m for all) and needs no assumption on the dependence between tests.

A detail specific to resampling p-values. With $B = 200$ placebo draws, the smallest possible rank p-value is $1/201 \approx 0.005$. With $m = 23$ tests, Holm requires $p_{(1)} \leq 0.05/23 \approx 0.0022$: no rank p-value can ever pass, even for a gap far outside the null distribution. The paper therefore applies Holm to a Gaussian-tail p-value, $p = 1 - \Phi(z)$ with $z = (\text{observed} - \text{null mean})/\text{null sd}$, which can be arbitrarily small. A result passes the verdict table only if both its rank p-value and its Holm-adjusted Gaussian p-value are below 5%. Nine of the 23 tests pass; two results that pass individually (the compensation slope on industries and the share of episodes where skewness falls) do not survive the adjustment.

8.10 Comparing nested forecasts: Clark–West

H7 compares a small model (volatility only, forecast f_1) with a larger one that adds an entropy regressor (forecast f_2). Under the null that the extra regressor is useless, its true coefficient is zero, but the larger model *estimates* it, which adds noise: f_2 has a slightly larger mean squared error even when the regressor is irrelevant. The Diebold–Mariano test on $d_t = (y_t - f_{1,t})^2 - (y_t - f_{2,t})^2$ is therefore biased against the larger model and under-rejects. Clark and West (2007) add back the noise term:

$$\tilde{d}_t = (y_t - f_{1,t})^2 - [(y_t - f_{2,t})^2 - (f_{1,t} - f_{2,t})^2],$$

and test $\mathbb{E}[\tilde{d}_t] = 0$ against $\mathbb{E}[\tilde{d}_t] > 0$ with a HAC standard error. Why this works: if the extra regressor is useless, f_2 equals f_1 plus pure estimation noise ν_t uncorrelated with the forecast error, so $\mathbb{E}[(y - f_2)^2] = \mathbb{E}[(y - f_1)^2] + \mathbb{E}[\nu^2]$. The DM differential then has mean $-\mathbb{E}[\nu^2] < 0$, and $(f_1 - f_2)^2 = \nu^2$ is exactly the term that re-centres it at zero. The test is one-sided and approximately normal.

Where this is used in the paper

Verdicts (tables 4 and 5 of the journal version): coupling under stress passes both nulls on both panels, also within years; crisis composition (H8), the range and daily half-life of $\mathcal{J}$ and the largest episode pass the i.i.d. null. One-sided jumps, the shape deficit under stress, the tails–correlation link (H6) and the daily sign (H5) are produced just as strongly, or more, by the null.

9 The geometry of stress scenarios

9.1 The divergence between two arbitrary Gaussians

Proposition 9.1. *For $p_1 = \mathcal{N}(\mu_1, \Sigma_1)$, $p_0 = \mathcal{N}(\mu_0, \Sigma_0)$ and $\delta = \mu_1 - \mu_0$,*

$$D_{KL}(p_1 \,\|\, p_0) = \tfrac{1}{2}\Big[\underbrace{\delta^\top \Sigma_0^{-1}\delta}_{\textit{shift}} + \underbrace{\operatorname{tr}(\Sigma_0^{-1}\Sigma_1) - N - \log\det(\Sigma_0^{-1}\Sigma_1)}_{\textit{deformation}}\Big].$$

Proof. $D_{KL}(p_1 \,\|\, p_0) = \mathbb{E}_{p_1}[\log p_1] - \mathbb{E}_{p_1}[\log p_0]$. The first term is $-h(p_1) = -\frac{1}{2}[N\log 2\pi + \log\det\Sigma_1 + N]$. For the second, the trace trick gives $\mathbb{E}_{p_1}[(X-\mu_0)^\top\Sigma_0^{-1}(X-\mu_0)] = \operatorname{tr}(\Sigma_0^{-1}\Sigma_1) + \delta^\top\Sigma_0^{-1}\delta$, so $-\mathbb{E}_{p_1}[\log p_0] = \frac{1}{2}[N\log 2\pi + \log\det\Sigma_0 + \operatorname{tr}(\Sigma_0^{-1}\Sigma_1) + \delta^\top\Sigma_0^{-1}\delta]$. Add them and use $\log\det\Sigma_0 - \log\det\Sigma_1 = -\log\det(\Sigma_0^{-1}\Sigma_1)$. □

Spectral reading of the deformation. If $\nu_1, \dots, \nu_N$ are the (positive) eigenvalues of $\Sigma_0^{-1}\Sigma_1$,

$$\text{deformation} = \tfrac{1}{2}\sum_k (\nu_k - 1 - \log\nu_k).$$

Each term is ≥ 0 (since $\log\nu \leq \nu - 1$) and zero if and only if $\nu_k = 1$: every deformed eigendirection costs something.

In one dimension. For a portfolio with mean m_0 and standard deviation s_0 under p_0, stressed to (m, s), let $u = (m - m_0)/s_0$ (shift in standard deviations) and $v = s/s_0$ (volatility multiplier). Then

$$g(u, v) = \tfrac{1}{2}(u^2 + v^2 - 1 - 2\log v). \tag{3}$$

9.2 The entropy-ball scenario

Definition 9.2. Given the current law p_0, a risk measure ρ and a radius η, the scenario of severity η is $p_\eta^\star = \arg\max\{\rho(p) : \; D_{KL}(p \,\|\, p_0) \leq \eta\}$: the worst law within η nats of the current one.

For a Gaussian loss, VaR and expected shortfall have the form $\rho = -m + c\,s$. For the 99 % ES, $c = \varphi(z_{0.99})/0.01 = \varphi(2.326)/0.01 \approx 2.665$.

Proposition 9.3 (Solution)**.** *The maximum of $-m + cs$ subject to $g(u, v) \leq \eta$ is reached on the boundary $g = \eta$, at a point such that*

$$u = -t, \qquad v - \frac{1}{v} = c\,t, \qquad t > 0.$$

For $c = 0$ (purely directional shock): $v = 1$ and $u = -\sqrt{2\eta}$.

Proof. Step 1: convexity. g is the sum of $u^2/2$ (convex) and $(v^2 - 1 - 2\log v)/2$, whose second derivative $1 + 1/v^2$ is positive: g is convex on $\mathbb{R} \times (0, \infty)$. The objective $s_0(-u + cv)$ (up to a constant) is linear. We maximise a linear function over a convex set: the Karush–Kuhn–Tucker conditions are necessary and sufficient.

Step 2: the constraint binds. A non-constant linear function has no interior maximum; the optimum lies on $g = \eta$.

Step 3: the Lagrangian. $\mathcal{L} = -u + cv - \lambda(g(u, v) - \eta)$. The first-order conditions are $-1 = \lambda u$ and $c = \lambda(v - 1/v)$. With $t = 1/\lambda > 0$: $u = -t$ and $v - 1/v = ct$. The scalar t is pinned down by $g(-t, v(t)) = \eta$.

Step 4: case $c = 0$. $v - 1/v = 0$ and $v > 0$ force $v = 1$, then $g = u^2/2 = \eta$ gives $u = -\sqrt{2\eta}$. □

Example 9.4 (Checking a row of the severity ladder (online appendix))**.** Ten-year rung: $u = -0.38$, $v = 1.63$. We check $g = \frac{1}{2}(0.144 + 2.657 - 1 - 0.977) = 0.412 \approx \eta = 0.408$, and $v - 1/v = 1.017 \approx c\,t = 2.665 \times 0.38 = 1.013$. With annual volatility of 20 %, that is $s_0 = 0.2/\sqrt{252} = 1.26\,\%$ per day, ES goes from $c\,s_0 = 3.4\,\%$ to $(0.38 + 2.665 \times 1.63)\,s_0 = 5.9\,\%$. The scenario spends its budget mostly on volatility (v) rather than direction (u), because ES is more sensitive to scale ($c \approx 2.7$) than to location (coefficient 1).

9.3 Calibrating the radius

We measure how much the market itself revises its law over $h = 21$ days: $I_t^{(21)} = D_{KL}(p_t^w \,\|\, p_{t-21}^w) = g(u_t, v_t)$, computed on the portfolio, with u_t, v_t the shift and volatility multiplier observed between $t-21$ and t. The history is cut into non-overlapping 21-day blocks (about 12 per year); a "ten-year" radius is the quantile exceeded on average once every 120 blocks, that is the $1 - 1/120$ quantile. Limit: compared with the history of volatility increases, this radius is about twice too mild; only the *relative* comparisons that follow are robust.

9.4 From the portfolio to all assets

Proposition 9.5 (Data-processing inequality)**.** *For any function f, if p^f and p_0^f are the laws of $f(X)$ under p and p_0, then $D_{KL}\left(p^f \,\|\, p_0^f\right) \le D_{KL}(p \,\|\, p_0)$.*

Proof sketch. Let $Y = f(X)$. The chain rule gives $D_{KL}(p_{X,Y} \,\|\, p_{0,X,Y}) = D_{KL}\left(p^f \,\|\, p_0^f\right) + \mathbb{E}_Y\left[D_{KL}\left(p_{X|Y} \,\|\, p_{0,X|Y}\right)\right]$. The second term is ≥ 0. Applying the same rule in the other order, $D_{KL}(p_{X,Y} \,\|\, p_{0,X,Y}) = D_{KL}(p \,\|\, p_0) + \mathbb{E}_X\left[D_{KL}\left(p_{Y|X} \,\|\, p_{0,Y|X}\right)\right]$, and this last term is zero because, $Y = f(X)$ being deterministic given X, the two conditional laws coincide. Hence $D_{KL}(p \,\|\, p_0) \ge D_{KL}\left(p^f \,\|\, p_0^f\right)$. □

In plain words

By looking at less (only the portfolio $w^\top X$), you cannot see more difference. The cost of a scenario on all assets is always at least its cost "seen by the portfolio". Table 7 of the journal version confirms it: 9.97 nats on 20 assets against 7.24 seen by the portfolio.

Proposition 9.6 (Minimal lift)**.** *Let $p_0 = \mathcal{N}(\mu_0, \Sigma_0)$, $s_0^2 = w^\top \Sigma_0 w$ and $\beta = \Sigma_0 w / s_0^2$ (the vector of asset betas on the portfolio). Among Gaussians that give the portfolio mean $m_0 + u s_0$ and standard deviation $v s_0$, the cheapest is*

$$\mu_1 = \mu_0 + u\, s_0\, \beta, \qquad \Sigma_1 = \Sigma_0 + (v^2 - 1)\, s_0^2\, \beta\beta^\top,$$

and its cost is exactly $g(u, v)$.

Proof. By proposition 9.5, every candidate costs at least $g(u, v)$. It is enough to show that this candidate attains the bound. We have $w^\top \beta = 1$ and $\beta^\top \Sigma_0^{-1} \beta = w^\top \Sigma_0 w / s_0^4 = 1/s_0^2$. Hence: $\delta^\top \Sigma_0^{-1} \delta = u^2 s_0^2 \cdot (1/s_0^2) = u^2$; $\operatorname{tr}(\Sigma_0^{-1}\Sigma_1) = N + (v^2 - 1) s_0^2\, \beta^\top \Sigma_0^{-1} \beta = N + v^2 - 1$; and, by the matrix determinant lemma ($\det(I + ab^\top) = 1 + b^\top a$), $\det(\Sigma_0^{-1}\Sigma_1) = 1 + (v^2 - 1) s_0^2\, \beta^\top \Sigma_0^{-1} \beta = v^2$. In proposition 9.1: $\frac{1}{2}(u^2 + v^2 - 1 - 2\log v) = g(u, v)$. □

Reading. Write $X = \beta f + e$ with $f = w^\top X$ the portfolio factor and e uncorrelated with f. The minimal scenario touches *only* the factor: it shifts its mean by u standard deviations and multiplies its volatility by v, leaving e intact. High-beta assets take the shock, and their correlations rise. **The least improbable scenario is a one-factor scenario.**

Proposition 9.7 (Cost of uniform dilation)**.** *If all volatilities are multiplied by v with correlations unchanged ($\Sigma_1 = v^2 \Sigma_0$), the deformation is $N \cdot \frac{1}{2}(v^2 - 1 - 2\log v)$: N times that of the minimal lift.*

Proof. Every eigenvalue of $\Sigma_0^{-1}\Sigma_1 = v^2 I$ equals v^2; we sum N identical terms $\frac{1}{2}(v^2 - 1 - \log v^2)$. □

Numerically, for $v = 1.63$ and $N = 20$: 0.34 nat per direction, so 6.80 nats in total, against 0.34 for the minimal lift. This is the "default: uniform dilation" row of the composition example in section 6.5 of the journal version.

9.5 Composing a scenario: volatility or correlation?

Proposition 9.8. *If $w^\top \Sigma_a w = w^\top \Sigma_b w = s^2$, then for every $\alpha \in [0, 1]$, $\Sigma_\alpha = (1 - \alpha)\Sigma_a + \alpha \Sigma_b$ satisfies $w^\top \Sigma_\alpha w = s^2$.*

Proof. $\Sigma \mapsto w^\top \Sigma w$ is linear. □

One can therefore mix an "all-volatility" covariance and an "all-correlation" covariance without changing portfolio severity; α is the *dependence share*. Since $\Sigma_1 \mapsto D_{KL}(\mathcal{N}(\mu, \Sigma_1) \,\|\, p_0)$ is convex (linear trace, convex $-\log\det$), the cost of a mixture is at most the mixture of the costs.

A ceiling for correlation. With volatilities fixed, $w^\top \Sigma w = \sum_{i,j} w_i w_j \sigma_i \sigma_j \rho_{ij}$ is largest when every ρ_{ij} equals 1, and then equals $(\sum_i w_i \sigma_i)^2$. On the main panel, this is only about 1.5 times current volatility: beyond that, volatilities must also be dilated.

Calibrating α. H1 gives the share of dependence in the move of Gaussian entropy under stress: $0.99/(0.99 + 5.97) \approx 14\,\%$. H8 refines it to $49\,\%$ over the most severe episodes. The larger α, the closer we get to the one-factor scenario, and the lower the cost (6.80 → 2.88 → 0.71 nats): proposition 9.7 makes this result expected rather than surprising.

9.6 Pricing a published scenario: the EBA example

A published scenario sets shocks on some variables S only (equities, long rates), not on the others U. The minimal cost is obtained by letting the unshocked variables take their most likely value.

Proposition 9.9 (Cost of a partial shock)**.** *For $p_0 = \mathcal{N}(0, \Sigma)$ and a shift $\delta = (\delta_S, \delta_U)$ of which only δ_S is imposed,*

$$\min_{\delta_U} \ \tfrac{1}{2}\delta^\top \Sigma^{-1}\delta = \tfrac{1}{2}\,\delta_S^\top \Sigma_{SS}^{-1}\delta_S, \qquad \text{reached at } \delta_U^\star = \Sigma_{US}\Sigma_{SS}^{-1}\delta_S.$$

Proof. Let $P = \Sigma^{-1}$, partitioned into blocks. The function $\delta_U \mapsto \delta^\top P\delta$ is strictly convex quadratic; its derivative vanishes at $\delta_U^\star = -P_{UU}^{-1}P_{US}\delta_S$, and the minimum is $\delta_S^\top(P_{SS} - P_{SU}P_{UU}^{-1}P_{US})\delta_S$. The matrix in brackets is the Schur complement of P_{UU} in P, and the block inversion formula says precisely that it equals Σ_{SS}^{-1}. Finally $-P_{UU}^{-1}P_{US} = \Sigma_{US}\Sigma_{SS}^{-1}$ (same formula): this is the Gaussian conditional expectation of X_U given $X_S = \delta_S$. □

Example 9.10 (Why correlation makes a shock cheaper)**.** Two indices with the same volatility, correlated at ρ, each fall by k standard deviations. With $\delta = (k, k)$ and $R^{-1} = \frac{1}{1-\rho^2}\begin{pmatrix} 1 & -\rho \\ -\rho & 1 \end{pmatrix}$,

$$\tfrac{1}{2}\delta^\top R^{-1}\delta = \frac{k^2}{1+\rho},$$

whereas the sum of the costs taken one by one is k^2. With $\rho = 0.87$ (average correlation of European indices), the joint cost is only 53 % of the sum. This is the mechanism that takes the EBA scenario from 31.9 nats (variable by variable) to 10.5 nats (joint): a common shock to highly correlated indices follows the most likely direction in the data.

The EBA figures. For the DAX alone, $\Delta = \log 0.5 = -0.693$ and $\sigma = 20.3\,\%$ per year: the shock is 3.41 standard deviations and costs $\frac{1}{2}(3.41)^2 = 5.8$ nats. The full scenario costs 10.5 nats, that is $\sqrt{2 \times 10.5} = 4.6$ standard deviations on a single factor.

Exercise 9.11. Show that the cost of a k-standard-deviation fall in N indices with constant correlation ρ is $\frac{Nk^2}{2(1+(N-1)\rho)}$.

Solution. $\delta = k\mathbf{1}$ is an eigenvector of R with eigenvalue $1 + (N-1)\rho$, so $\delta^\top R^{-1}\delta = k^2 N/(1 + (N - 1)\rho)$.

9.7 Reverse stress testing

We reverse the question: what is the cheapest scenario that produces a given loss L? In the Gaussian setting, it is proposition 9.3 read backwards. In the non-parametric setting, the solution is an *exponential tilting* of the sample: observed days are reweighted with weights proportional to $e^{\theta\,\mathrm{loss}_t}$. As θ grows, almost all the weight goes to a few days; the effective sample size $(\sum_t \omega_t)^2 / \sum_t \omega_t^2$ collapses (it is 1.0 at 9 nats on the main panel), and one must go back to the Gaussian version.

10 Decomposition by segment: the entropic CISS

Split the N series into blocks $B_1, \dots, B_K$ (money market, bonds, equities, intermediaries, FX). Let R_{kk} be the diagonal blocks of R and $R_D = \operatorname{diag}(R_{11}, \dots, R_{KK})$ the block-diagonal matrix.

Proposition 10.1.

$$\underbrace{C(R)}_{\text{total}} = \sum_{k=1}^{K} \underbrace{C(R_{kk})}_{\text{within block } k} + \underbrace{\left(-\tfrac{1}{2}\log\det(R_D^{-1/2} R R_D^{-1/2})\right)}_{\text{between blocks}},$$

and each of the $K+1$ terms is non-negative. The between-block term is the Gaussian mutual information between the blocks; it is zero if and only if the blocks are pairwise uncorrelated.

Proof. Identity. $R = R_D^{1/2}\,\tilde{R}\,R_D^{1/2}$ with $\tilde{R} = R_D^{-1/2} R R_D^{-1/2}$. Since the determinant is multiplicative, $\log\det R = \log\det R_D + \log\det \tilde{R}$, and $\log\det R_D = \sum_k \log\det R_{kk}$ (determinant of a block-diagonal matrix). Multiply by $-\frac{1}{2}$.

Signs. Each R_{kk} is a correlation matrix, so $C(R_{kk}) \geq 0$ by proposition 4.2. The matrix $\tilde{R}$ has identity diagonal blocks, hence a diagonal of 1s: it is again a correlation matrix, and the same argument applies. (Equivalently, $\det R \leq \prod_k \det R_{kk}$ is Fischer's inequality.) □

Two blocks: canonical correlations. For $K = 2$, let $M = R_{11}^{-1/2} R_{12} R_{22}^{-1/2}$; its singular values $\rho_1, \dots, \rho_r$ are the *canonical correlations* between the two segments. We have $\tilde{R} = \begin{pmatrix} I & M \\ M^\top & I \end{pmatrix}$ and, by the Schur complement formula, $\det \tilde{R} = \det(I - MM^\top) = \prod_j (1 - \rho_j^2)$. Hence

$$I(B_1; B_2) = -\tfrac{1}{2} \sum_j \log(1 - \rho_j^2)\,:$$

the sum of the information carried by each canonical pair, each computed as in proposition 5.2.

In plain words

Total coupling splits exactly into "what happens inside each market" and "what ties markets together". The second term is the systemic part. Unlike the CISS, which aggregates, this decomposition says *which* segments stress ties together, and by how much, in a single unit.

The rank version. Replace each series by $\Phi^{-1}(\hat{F}_i(x_i))$, where $\hat{F}_i$ is its empirical distribution function: this gives "normal scores". A monotone transformation of the marginals does not change ranks, so the measure becomes insensitive to each series' fat tails: it estimates the multi-information of a *Gaussian copula*. This is the most robust version against the null (9.6 and 12.7 standard deviations).

Link with the structural index. With the chain rule,

$$\mathcal{J} \approx \underbrace{\textstyle\sum_i \mathcal{D}_i}_{\text{shape}} + \underbrace{\textstyle\sum_k C(R_{kk})}_{\text{within}} + \underbrace{\left(-\tfrac{1}{2}\log\det \tilde{R}\right)}_{\text{between}}.$$

The entropic CISS keeps the last two terms and drops shape, which is what the null reproduces. It is the structural index stripped of its non-robust channel.

11 Summary: each formula and its role

Result	Role in the paper
$h(aX) = h(X) + \log \lvert a \rvert$	the level of entropy has no absolute meaning; we work with divergences
$D_{KL}(p \,\|\, q) \geq 0$, invariant under change of variables	common unit across all asset classes
$\frac{1}{2}\Delta^\top \Sigma^{-1} \Delta$; $k\sigma \leftrightarrow k^2/2$ nats	nats–standard deviations exchange rate; scenario audit
Gaussian maximum entropy	the Gaussian is the zero point
$h(q) = \frac{N}{2}\log(2\pi e) + \sum \log \sigma_i + \frac{1}{2}\log\det R$	separating scale and dependence
$\det R \leq 1$	coupling C is non-negative
$\mathcal{D} = h(q) - h(p)$; $\mathcal{J} = \mathcal{D} + C = D_{KL}(p \,\|\, \prod q_i)$	the structural index is a true divergence
$\mathbb{E}_Y[h(X \mid Y)] = h(X) - I(X;Y)$	upward jumps of $\mathcal{J}$ (theorem)
OU, $t_{1/2} = \ln 2/\kappa$	relaxation (postulate), half-life
Jensen on a convex estimator	upward bias; need for a null
Bartlett: $\mathbb{E}[C(\hat{R})] \approx N(N-1)/(4T')$	size of the bias; compare gaps, not levels
Matrix determinant lemma, one common day	the mechanical jump of coupling
Exchangeability, $(1+k)/(B+1)$	exact resampling p-values
CCC-GARCH filtered historical simulation	null that keeps volatility, not correlation, dynamics
Year fixed effects (Frisch–Waugh–Lovell)	stress effect within a year, not a trend
Holm step-down	family-wise error control in the verdict table
Clark–West	forecast test for nested models (H7)
KL of two Gaussians, $g(u, v)$	cost of a scenario on a portfolio
KKT: $u = -t$, $v - 1/v = ct$	entropy-ball scenario
Data processing; minimal lift	the minimal scenario is one-factor
Uniform dilation $= N\times$ minimum	why classical scenarios are too expensive
Schur complement	cost of a partial shock (EBA)
Within / between-block decomposition	entropic CISS, contagion map

Further reading. Cover and Thomas, *Elements of Information Theory* (2006), chapters 2, 8 and 12, cover entropy, divergence, differential entropy and maximum entropy. Breuer and Csiszár (2013) and Glasserman and Xu (2014) develop Kullback–Leibler ball scenarios. The project's theoretical supplement gives the proofs in the paper's exact notation.